1 Cooperative diffusivity from the displacement transients

In one dimension the solvent mass balance, Darcy's law and the Hookean network stress $\sigma'_{zz} = M\,\partial u_z/\partial z$ combine to

$$\frac{\partial u_z}{\partial t} = q(t) + \mathcal{D}_c \frac{\partial^2 u_z}{\partial z^2}, \qquad \mathcal{D}_c \equiv \kappa M, \tag{1}$$

where $q(t) = \phi_s v_s + \phi_p v_p$ is the total volumetric flux, uniform in z. The strain $\varepsilon \equiv -\partial u_z/\partial z$ then obeys

$$\frac{\partial \varepsilon}{\partial t} = \mathcal{D}_c \frac{\partial^2 \varepsilon}{\partial z^2} \tag{2}$$

on $0 \le z \le L$, with L the thickness of the configuration the modes are written on and $\zeta \equiv z/L$. Each protocol below fixes the boundary and initial conditions of (2); the fitted quantity is the decay rate of the first mode, τ_1^{-1}, so $\mathcal{D}_c \propto L^2$ for a given τ_1.

1.1 Compression, loading

After the ramp both plates are held, $u_z(0,t) = 0$ and $u_z(L,t) = -\Delta L$, and both faces drain into the same bath, so the strain is periodic:

$$\varepsilon(0,t) = \varepsilon(L,t), \qquad \left.\frac{\partial \varepsilon}{\partial z}\right|_0 = \left.\frac{\partial \varepsilon}{\partial z}\right|_L. \tag{3}$$

The solution of (2) and (3) is

$$\varepsilon(\zeta,t) = \frac{\Delta L}{L} + \sum_{m=1}^{\infty} \left[a_m \cos(2m\pi\zeta) + b_m \sin(2m\pi\zeta)\right] e^{-4m^2\pi^2\mathcal{D}_c t/L^2}, \qquad \tau_1 = \frac{L^2}{4\pi^2\mathcal{D}_c}, \tag{4}$$

with a_m, b_m set by the state at the end of the ramp. Integrating (4) gives the displacement modes $\sin(2m\pi\zeta)$ and $\cos(2m\pi\zeta) - 1$, both zero at both plates.

1.2 Compression, unloading

At $t = 0$ the plates are withdrawn from a gel of uniform strain $\varepsilon_0 = \Delta L/L$. Both faces are stress free,

$$\varepsilon(0,t) = \varepsilon(L,t) = 0, \qquad \varepsilon(\zeta,0) = \varepsilon_0, \tag{5}$$

so

$$\varepsilon(\zeta,t) = \varepsilon_0 \sum_{k\ \text{odd}} \frac{4}{k\pi} \sin(k\pi\zeta)\, e^{-k^2\pi^2\mathcal{D}_c t/L^2}, \qquad \tau_1 = \frac{L^2}{\pi^2\mathcal{D}_c}, \tag{6}$$

and the thickness recovers as

$$\frac{\delta L(t)}{\delta L(\infty)} = 1 - \sum_{k\ \text{odd}} \frac{8}{(k\pi)^2}\, e^{-k^2\pi^2\mathcal{D}_c t/L^2}. \tag{7}$$

The displacement modes are $\cos(k\pi\zeta)$ in the frame of the compressed gel. The uniform mode of (4) is absent here and the $m = 1$ rate of (4) is four times the $k = 1$ rate of (6): at equal $\mathcal{D}_c$ the unload is four times slower than the hold.

1.3 Permeation, approach to steady state

The network is pinned at the support, $u_z(0,t) = 0$, and the feed face is free. The pore pressure drop $\Delta P(t)$ is carried by the network at the support and vanishes at the feed face:

$$\varepsilon(0,t) = \frac{\Delta P(t)}{M}, \qquad \varepsilon(L,t) = 0, \qquad \varepsilon(\zeta,0) = 0. \tag{8}$$

The steady state of (1) under (8) is the planar profile,

$$u_z(\zeta,\infty) = -\frac{\Delta P L}{M}\left(\zeta - \frac{\zeta^2}{2}\right), \qquad u_F \equiv u_z(L,\infty) = -\frac{\Delta P L}{2M}, \tag{9}$$

which is (S127) with $z \to -z$. Writing the applied history as increments w_j at t_j, $\Delta P(t) = \Delta P \sum_{t_j<t} w_j$, $\sum_j w_j = 1$, the transient is

$$u_z(\zeta,t) = \sum_{k=1}^{\infty} A_k \left[1 - \cos(k\pi\zeta)\right] H_k(t), \qquad H_k(t) = \sum_{t_j<t} w_j \left[1 - e^{-k^2\pi^2 \mathcal{D}_c (t-t_j)/L^2}\right], \qquad \tau_1 = \frac{L^2}{\pi^2 \mathcal{D}_c}, \tag{10}$$

and the feed-face displacement follows

$$\frac{u_F(t)}{u_F(\infty)} = \sum_{k \text{ odd}} \frac{8}{(k\pi)^2} H_k(t). \tag{11}$$

1.4 Fitting

In each case the mode amplitudes are free and $\mathcal{D}_c$ is the single nonlinear parameter, found by least squares over the displacement snapshots of the first $6\tau_1$ (τ_1 of the fit itself, iterated), each mode decay averaged over the output window of the snapshot it is compared with. The thickness traces (7) and (11) give a second $\mathcal{D}_c$ with one amplitude, $\delta L(\infty)$ or $u_F(\infty)$. Then $M = \Delta P L/(2|u_F|)$ from (9) and $\kappa = \mathcal{D}_c/M$.